UrbanStyle Apparel

Fashion Pricing Analysis

1. Project Overview

This is a self-led end-to-end analytics project that analyses the pricing structure, discount behaviour, and brand profitability of men's t-shirts listed on a major Indian e-commerce platform.

The dataset contains 1,428 SKUs across 143 brands. The data was first cleaned in Azure SQL, then connected to Power BI Desktop for modelling and visualisation, and finally published to Power BI Service for live interactive access.

The central observation that motivated this project: Indian e-commerce platforms display massive discount percentages — 60%, 70%, even 89% off. But when you look at the actual numbers, MRP is often set 2x, 3x, or even 9x above the real selling price. The discount is manufactured. This project was built to prove that with data.

Business Question

How can catalog pricing data be used to identify discount manipulation, understand tier-wise profitability, and help a retail business make better decisions about catalog strategy, brand partnerships, and pricing guardrails?

2. Tools & Technology Stack

Tool / Platform	How It Was Used
Azure SQL (Cloud Database)	Uploaded raw dataset. Cleaned price columns using SQL UPDATE queries. Both Sale Price and MRP had special characters that were removed using TRIM and REPLACE before connecting to Power BI.
Power BI Desktop	Connected directly to Azure SQL. Built star schema data model. Created calculated columns and 12 DAX measures. Designed 3 report pages with charts, KPI cards, and slicer interactions.
Power BI Service	Published the completed report for live interactive access. Viewers can use slicers and filters without downloading any file.
DAX (Data Analysis Expressions)	Written inside Power BI for all 12 measures including AVERAGEX, MAXX, MINX, SUMMARIZE, TOPN, FILTER, DIVIDE, and COUNTROWS.
SQL (Azure Query Editor)	Used UPDATE, SET, TRIM, REPLACE, CAST, LIKE for data cleaning directly in the cloud database.

3. Data Cleaning in Azure SQL

Before building the dashboard, the raw data was uploaded to an Azure SQL database (testserverpowerbi / Test). The table dbo.[Men Tshirt] contained 1,428 rows with four columns: Brand, Title, Original_Price, and Sale_Price.

Both price columns were stored as text (nvarchar) and contained a special character '?' embedded in the values. This had to be removed before the prices could be used as numbers in Power BI calculations.

Cleaning Steps Performed

Step 1 — Remove '?' from Original_Price (MRP)

```
UPDATE [dbo].[Men Tshirt]
SET original_price = TRIM(REPLACE(CAST(original_price AS varchar(max)), '?', ''))
WHERE original_price LIKE '%?%'
```

Result: 1,428 rows affected. All '?' characters removed from the MRP column.

Step 2 — Remove '?' from Sale_Price

```
UPDATE [dbo].[Men Tshirt]
SET sale_price = TRIM(REPLACE(CAST(sale_price AS varchar(max)), '?', ''))
WHERE sale_price LIKE '%?%'
```

Result: 1,428 rows affected. All '?' characters removed from the Sale Price column.

After cleaning, both columns contained clean numeric values as text, which Power BI then converted to decimal numbers during the data connection step. This cleaning was essential — without it, Power BI would read the prices as text and all calculations would fail.

Column	Before Cleaning	After Cleaning
Original_Price (MRP)	1,749? or ?1,749	1749
Sale_Price	525? or ?525	525
Data Type	Text (nvarchar)	Text (nvarchar) — converted to Decimal in Power BI
Rows Affected	1,428	1,428

4. Data Model

The model uses a Star Schema — a Fact table connected to a Dimension table via the Brand column. This design keeps DAX calculations clean, context transitions correct, and avoids circular dependencies.

Table	Type	Description
T-Shirt Products	Fact Table — 1,428 rows	One row per SKU. Contains Brand, Title, Sale Price, MRP, Cost Price, and all calculated columns.
Brand Tier Reference	Dimension Table — 143 rows	One row per brand. Contains Tier, Origin, and Type (Domestic or International).

Relationship: T-Shirt Products[Brand] → Brand Tier Reference[Brand] (Many-to-One)

Calculated Columns Created in Power BI

Column	Table	Formula Logic	Purpose
Profit Per Unit	T-Shirt Products	Sale Price - Cost Price	Absolute profit on each SKU
Profit Margin %	T-Shirt Products	Profit Per Unit / Sale Price	Margin % of sale price
Discount %	T-Shirt Products	(MRP - Sale Price) / MRP	Actual discount on MRP
Discount Amount	T-Shirt Products	MRP - Sale Price	Rupee discount value
Price Band	T-Shirt Products	IF Sale Price < 500 = Economy, < 1000 = Budget, < 2000 = Mid-range, < 5000 = Premium, else Luxury	Groups SKUs into price buckets
Discount Flag	T-Shirt Products	IF Discount % >= 70% = High Discount, else Normal	Flags extreme discounting

5. DAX Measures

12 measures were written and organised into 4 display folders inside Power BI Desktop. All measures are stored in the T-Shirt Products table.

Folder 1 — Core Metrics

Measure	Formula	Page Used	Purpose
Total Products	COUNTROWS('T-Shirt Products')	Page 1, 3	Count of SKUs in filter context
Total Brands	DISTINCTCOUNT('T-Shirt Products'[Brand])	Page 1, 3	Count of unique brands
Avg Sale Price	AVERAGE('T-Shirt Products'[Sale Price])	Page 1, 3	Average sale price benchmark

Avg Discount %	AVERAGEX('T-Shirt Products', DIVIDE(MRP - Sale Price, MRP))	Page 1, 2	Row-by-row avg discount — more accurate than column average
MRP Inflation Ratio	DIVIDE(AVERAGE(MRP), AVERAGE(Sale Price))	Page 1	How many times MRP exceeds actual sale price

Folder 2 — Profitability

Measure	Formula	Page Used	Purpose
Avg Profit Per Unit	AVERAGEX('T-Shirt Products', Sale Price - Cost Price)	Page 2, 3	Average profit per SKU — row-by-row for accuracy
Est Avg Profit Margin %	AVERAGE('T-Shirt Products'[Profit Margin %])	Page 2	Overall margin health of filtered catalog

Folder 3 — Discount Analysis

Measure	Formula	Page Used	Purpose
Extreme Discount %	DIVIDE(COUNTROWS(FILTER(..., Discount >= 70%)), COUNTROWS(...))	Page 2	% of SKUs with 70%+ discount — fake anchor pricing measure
High Discount Product Count	COUNTROWS(FILTER(..., Discount Flag = 'High Discount'))	Page 2	Absolute count of High Discount SKUs

Folder 4 — Brand Intelligence

Measure	Formula	Page Used	Purpose
Count of Brands	DISTINCTCOUNT('Brand tier reference'[Brand])	Page 3	Distinct brand count from dimension table
Top Brand	MAXX(TOPN(1, SUMMARIZE(..., SKU Count), DESC), Brand)	Page 3	Brand with most SKUs in filter context
Highest Avg Price by Brand	MAXX(SUMMARIZE(..., Avg Sale Price by Brand), [Avg Price])	Page 3	Highest brand-level avg sale price
Lowest Avg Price by Brand	MINX(SUMMARIZE(..., Avg Sale Price by Brand), [Avg Price])	Page 3	Lowest brand-level avg sale price

6. Dashboard Pages

Page 1 — Catalog Overview

Subtitle: Catalog health | Pricing dynamics | Brand assortment depth

Slicers: Tier, Price Band, Type

KPI Cards

KPI Card	Value	Measure Used
Total Products	1,428	Total Products
Total Brands	143	Total Brands
Avg Sale Price	Rs.1,034	Avg Sale Price
Avg Discount %	57.1%	Avg Discount %
MRP Inflation Ratio	2.32x	MRP Inflation Ratio

Charts

Chart	What It Shows
Brand Pricing Dynamics Scatter	Each dot = one brand, size = SKU count, colour = tier. Shows how MRP sits above actual sale price. Brands far above the diagonal line have highly inflated MRP.
Top 15 Brands by SKU Count	Ranked bar chart. The Indian Garage Co leads with 184 SKUs — 12.9% of the entire catalog.
MRP vs Sale Price vs Profit Per Unit by Tier	Grouped bars showing all three metrics side by side for each tier. Shows the gap between listed price and actual profit.
SKU Count and Avg Profit Per Unit by Tier	Bars = SKU volume per tier. Line = avg profit per unit. Budget has most SKUs (560) but lowest profit (Rs.99). Luxury fewest (58) but highest profit (Rs.1,586).

Key Insights on Page 1

- Budget tier has 560 SKUs but earns only Rs.99 profit per unit — Luxury earns Rs.1,586 from just 58 SKUs.
- 57% average discount is manufactured — MRP is artificially inflated 2.32x before discounting.
- International brands earn 4x more profit per unit than domestic brands despite fewer SKUs.

Page 2 — Discount & Profitability

Subtitle: Pricing behaviour | Discount depth | Brand margin analysis

KPI Cards

KPI Card	Value	Measure Used
Est Avg Profit Margin %	26.3%	Est Avg Profit Margin %
Avg Discount %	57.1%	Avg Discount %
Extreme Discount %	26.5%	Extreme Discount %
Avg Profit Per Unit	Rs.243	Avg Profit Per Unit

Charts

Chart	What It Shows
Avg Discount % by Brand (Top 10)	Ranked bars of most heavily discounting brands. iVOC leads at 89%. All top 10 exceed 79%.
Avg Discount % by Tier	Economy = 73.1%, Budget = 59.9%, Mid-range = 58.5%, Premium = 45.6%, Luxury = 45.5%. 27-point spread between Economy and Luxury.
Avg Discount % vs Avg Sale Price Scatter	High discount brands cluster at low sale price. Luxury and Premium sit at lower discount and higher price — shows pricing discipline.
Brand Discount & Profit Performance Table	Brand, Avg Profit Per Unit, Avg Discount %, and Discount Flag side by side. Most high-discount brands earn only Rs.57-Rs.150 per unit.

Key Insights on Page 2

- iVOC inflates MRP 9.1x — lists at Rs.3,499, sells at Rs.385. The 89% discount was never real.
- Economy tier averages 73% discount with 4.47x MRP inflation — highest manufactured pricing in the catalog.
- 26.5% of products carry 70%+ discount — 1 in 4 SKUs depends on fake anchor pricing.

Page 3 — Brand Intelligence

Subtitle: How do brands compare on price, profit and discount?

Slicers: Brand, Tier, Price Band

KPI Cards

KPI Card	Value	Measure Used
Count of Brands	143	Count of Brands
Top Brand	The Indian Garage Co	Top Brand
Lowest Avg Price by Brand	Rs.285	Lowest Avg Price by Brand
Highest Avg Price by Brand	Rs.6,086	Highest Avg Price by Brand

Charts

Chart	What It Shows
Brand Pricing & Profit Breakdown Table	Full brand comparison — Avg Sale Price, Avg MRP, Avg Discount %, SKU Count, Avg Profit Per Unit, Tier. ARMANI EXCHANGE earns Rs.2,282/unit.
Top 15 Brands by Avg Sale Price	ARMANI EXCHANGE leads at Rs.6,086. Colour = tier. All top 4 brands are Luxury or International Premium.
Avg Sale Price: Domestic vs International by Tier	Clustered bars comparing domestic vs international brands within each tier. International Luxury = Rs.4,761. Domestic Budget = Rs.551.

Key Insights on Page 3

- International brands earn Rs.575 profit per unit — 4x more than domestic brands at Rs.141 — despite being only 24% of SKUs.
- ARMANI EXCHANGE earns Rs.2,282 per unit after 48% discount — more than the entire Budget tier earns at full price (Rs.99/unit).
- The Indian Garage Co has 184 SKUs (12.9% of catalog) but earns only Rs.77 per unit — lowest among top 10 brands.

7. Key Business Findings

Finding 1 — MRP Inflation is Systemic, Not Occasional

The average MRP is 2.32x the actual sale price across all 1,428 SKUs. iVOC inflates MRP 9x. Economy tier inflates 4.47x. This is not a few outlier sellers — it is how large parts of the catalog are structured. The 57% average discount is not a sale. It is the pricing model.

Finding 2 — Volume Without Margin is a Business Trap

Budget tier holds 39% of all SKUs but earns only Rs.99 per unit. The Indian Garage Co has 184 SKUs — the most of any brand — but earns the least profit per unit among the top 10 brands (Rs.77). More products does not mean more profit.

Finding 3 — Luxury Tier is Underrepresented and Highly Profitable

Luxury has only 58 SKUs (4% of catalog) but earns Rs.1,586 per unit — 16x more than Budget. Adding 50 Luxury SKUs would generate more total profit than adding 800 Budget SKUs. The most profitable segment is the most underserved.

Finding 4 — International Brands Outperform on Every Margin Metric

International brands are 24% of SKUs but earn Rs.575 profit per unit on average — 4x more than domestic brands at Rs.141. They also discount less. Brand equity is a more powerful profit driver than volume.

Finding 5 — 1 in 4 Products Has a Fake Anchor Price

26.5% of SKUs carry 70%+ discounts. These are not genuine markdowns. MRP was set artificially high to manufacture a false sense of savings. This erodes consumer trust and makes genuinely priced products impossible to compete.

Finding 6 — Economy Tier Has No Clear Strategic Purpose

Economy tier has 73% average discount (highest of all tiers), Rs.96 profit per unit (second lowest), and only 141 SKUs (second fewest). It neither serves volume nor margin. It is the most inefficient tier in the catalog.

8. How Can This Be Improved — Recommendations

Recommendation	Business Impact
Expand Luxury and Premium Assortment	50 Luxury SKUs generate more profit than 800 Budget SKUs. Target international or aspirational domestic brands with strong brand equity.
Cap MRP Inflation Platform-Wide	Set a rule: MRP cannot exceed 3x sale price. Products with 70%+ discounts require approval. Protects consumer trust and platform credibility.
Reconsider Economy Tier Positioning	Merge Economy into Budget or remove low-margin SKUs. The tier adds complexity without adding value — high discount, low profit, low SKU count.
Invest in International Brand Acquisition	International brands earn 4x more per unit. 10-15 new international brands in Premium/Luxury would materially lift platform-wide average profit.
Reduce Catalog Concentration Risk	Indian Garage Co holds 12.9% of catalog. Diversifying across multiple mid-size brands reduces single-brand dependency risk.
Use Discount Flags in Search Rankings	Deprioritise 70%+ discount products in search. Surface genuinely priced products to rebuild consumer trust over time.

9. Future Improvements

Version 2.0

- Add actual units sold per SKU to calculate total revenue and total profit — not just per-unit averages. This is the most important missing piece.
- Add a date dimension to track how prices, MRP, and discounts change over time. Enable trend analysis — is MRP inflation increasing?
- Add a fourth report page focused on customer segments and which price bands generate the most revenue.

Version 3.0

- Expand beyond t-shirts to shirts, trousers, footwear, and women's apparel for cross-category comparison.
- Compare the same brands across Flipkart, Myntra, and Amazon — which platform has the most genuine discounting?
- Build a predictive pricing model to forecast whether a new listing is genuinely discounted or manufactured.

10. Current Limitations

Limitation	Explanation
Cost Price is Estimated	Profit calculations use an estimated cost price. Real cost varies by brand, quantity, and season. Profit figures are directional, not precise financials.
No Sales Volume Data	Dataset has listed SKUs but not actual units sold. SKU count is used as a volume proxy — not the same as transaction data.

Static Snapshot	Point-in-time dataset. E-commerce prices and discounts change frequently. Insights are valid for the collection period only.
Men's T-Shirts Only	Findings may not generalise to other categories like footwear, formal wear, or women's apparel.
Manual Tier Classification	Tiers assigned at brand level. A Premium brand may have individual SKUs priced in the Budget range, affecting tier averages.

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